

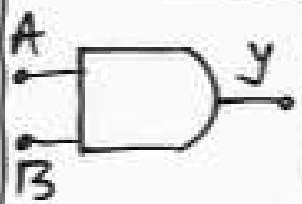
Gate	Symbol	Equation	Truth Table
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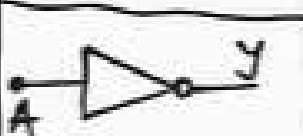
OR



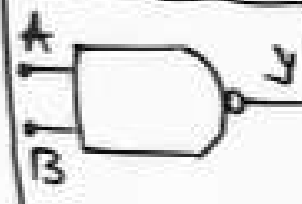
$y = A + B$

A	B	$y = A + B$
1	1	1
1	0	1
0	1	1
0	0	0

A AND B		$y = A \cdot B$	<table><tr><th>A</th><th>B</th><th>$y = A \cdot B$</th></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	A	B	$y = A \cdot B$	1	1	1	1	0	0	0	1	0	0	0	0
A	B	$y = A \cdot B$																
1	1	1																
1	0	0																
0	1	0																
0	0	0																

NOT		$y = \bar{A}$	<table><tr><th>A</th><th>$y = \bar{A}$</th></tr><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td></tr></table>	A	$y = \bar{A}$	1	0	0	1
A	$y = \bar{A}$								
1	0								
0	1								

NOR		$y = \overline{A+B}$	<table><tr><th>A</th><th>B</th><th>$y = \overline{A+B}$</th></tr><tr><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td></tr></table>	A	B	$y = \overline{A+B}$	1	1	0	1	0	0	0	1	0	0	0	1
A	B	$y = \overline{A+B}$																
1	1	0																
1	0	0																
0	1	0																
0	0	1																

NAND		$y = \overline{A \cdot B}$	<table><tr><th>A</th><th>B</th><th>$y = \overline{A \cdot B}$</th></tr><tr><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td></tr></table>	A	B	$y = \overline{A \cdot B}$	1	1	0	1	0	1	0	1	1	0	0	1
A	B	$y = \overline{A \cdot B}$																
1	1	0																
1	0	1																
0	1	1																
0	0	1																
XOR		$y = A \oplus B$	<table><tr><th>A</th><th>B</th><th>$y = A \oplus B$</th></tr><tr><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	A	B	$y = A \oplus B$	1	1	0	1	0	1	0	1	1	0	0	0
A	B	$y = A \oplus B$																
1	1	0																
1	0	1																
0	1	1																
0	0	0																

XOR		$Y = A\bar{B} + \bar{A}B$	<table border="1"><tr><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	1	1	0	1	0	1	0	1	1	0	0	0
1	1	0													
1	0	1													
0	1	1													
0	0	0													

XNOR		$Y = AB + \overline{AB}$	0 0 0
			1 0 0 0 1 0 0 0 1